

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-FA-8901-06 Sheet 1 of 1
OPL Title	Corrosion / Thickness Inspection	Discipline	Mechanical	
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	Area / Unit	8900 - HEXANE COLUMN OVERHEAD SYSTEM	
Related Interlock	SEQ-8901 (SIL 1)	P&ID; Ref	TJC-LLD-PID-8901	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **corrosion / thickness inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Take UT thickness readings on the shell bottom and boot	Isolation & LOTO verified
2	Compare readings to the 3.0 mm corrosion allowance	Parts match BOM / spec
3	Inspect the mist pad and vortex breaker at turnaround	Torque / clearance to spec
4	Record the readings in the RBI / inspection file	No leak / smooth operation
5	Flag any accelerated wall loss to inspection engineering	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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